

In-Class Activity-2

Design and Validate a Hybrid Network Using Cisco Packet Tracer

Objective

Design, configure, and validate a hybrid network topology by combining two different network infrastructures. Implement either an HTTP or FTP service and demonstrate successful end-to-end communication.

Problem Statement

Design a hybrid network consisting of **two different topologies** (e.g., Star + Bus, Star + Ring, Star + Mesh, Tree + Star, etc.) using Cisco Packet Tracer.

Your network must include:

- Minimum **2 different network topologies**
- At least **8 end devices (PCs/Laptops)**
- **2 Switches**
- **1 Router**
- One **Server** configured as either:
 - HTTP Server **or**
 - FTP Server
- Appropriate cabling

Tasks

1. Design a hybrid topology by combining two different network infrastructures.
2. Create an IP addressing scheme for all devices.
3. Configure IP addresses, subnet masks, and default gateways.
4. Configure the router interfaces.
5. Configure the server:
 - Enable **HTTP** service and create a sample webpage**OR**
 - Enable **FTP** service and create a user account.
6. Verify connectivity using:
 - ping
 - tracert
7. Validate the application layer:
 - Access the webpage through a browser (HTTP)**OR**
 - Successfully log in and transfer a file using FTP.

8. Save the Packet Tracer file.

Validation Requirements

Demonstrate the following during evaluation:

- ✓ All devices receive correct IP configuration.
- ✓ All hosts can communicate successfully.
- ✓ No packet loss during ping.
- ✓ HTTP webpage is accessible **or** FTP login and file transfer are successful.
- ✓ Network is logically organized and properly labeled.

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	GigabitEthernet0/0 (G0/0)	192.168.10.1	255.255.255.0	—
Router0	GigabitEthernet0/1 (G0/1)	192.168.20.1	255.255.255.0	—
Switch0	VLAN 1 (<i>Not Configured</i>)	—	—	—
Switch1	VLAN 1 (<i>Not Configured</i>)	—	—	—
PC0	FastEthernet0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	FastEthernet0	192.168.10.3	255.255.255.0	192.168.10.1
PC2	FastEthernet0	192.168.10.4	255.255.255.0	192.168.10.1
PC3	FastEthernet0	192.168.10.5	255.255.255.0	192.168.10.1
PC4	FastEthernet0	192.168.20.4	255.255.255.0	192.168.20.1
PC5	FastEthernet0	192.168.20.2	255.255.255.0	192.168.20.1
PC6	FastEthernet0	192.168.20.3	255.255.255.0	192.168.20.1
PC7	FastEthernet0	192.168.20.5	255.255.255.0	192.168.20.1
Server	FastEthernet0	192.168.20.10	255.255.255.0	192.168.20.1

Results:

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Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.9

Pinging 192.168.1.9 with 32 bytes of data:

Reply from 192.168.1.9: bytes=32 time=10ms TTL=128
Reply from 192.168.1.9: bytes=32 time<1ms TTL=128
Reply from 192.168.1.9: bytes=32 time=10ms TTL=128
Reply from 192.168.1.9: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.1.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 5ms

C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time=5ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time=13ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time=16ms TTL=128

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 7ms

C:\>http://192.168.1.100
Invalid Command.

C:\>ipconfig

FastEthernet0 Connection: (default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . .: FE80::C0A1:F05F:F82D:1D3
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Traceroute result:

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0.0.0.0

C:\>tracert 192.168.1.100

Tracing route to 192.168.1.100 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.1.100

Trace complete.

C:\>ftp 192.168.1.100

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ftp:

```

C:\>tracert 192.168.1.100

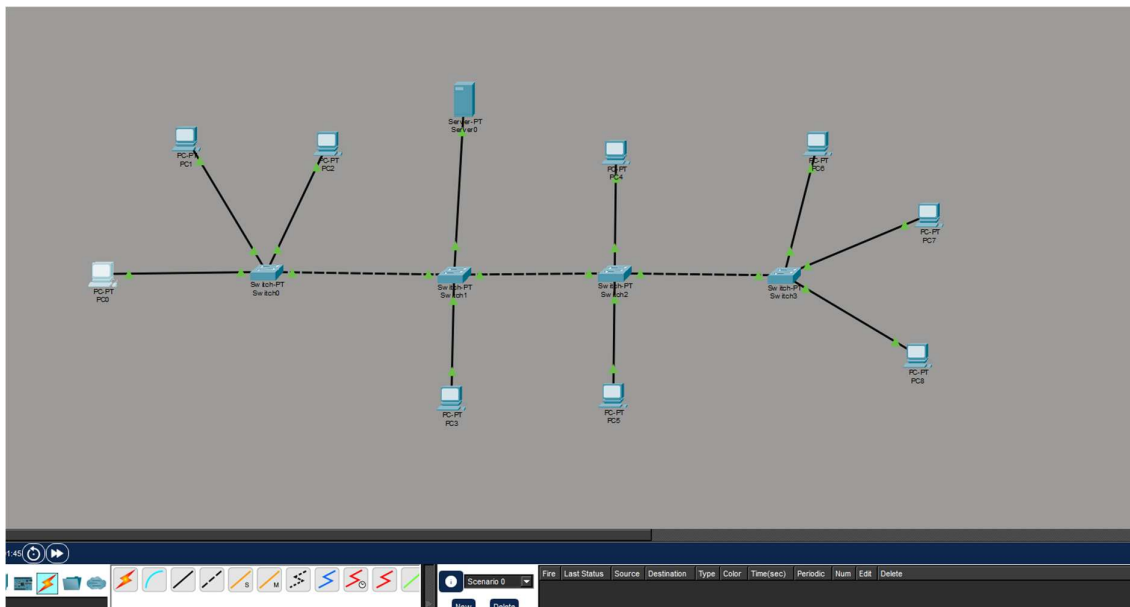
Tracing route to 192.168.1.100 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.1.100

Trace complete.

C:\>ftp 192.168.1.100
Trying to connect...192.168.1.100
Connected to 192.168.1.100
220- Welcome to PT Ftp server
Username:amrita
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>

```



HTTP webpage screenshot or FTP login screenshot:

